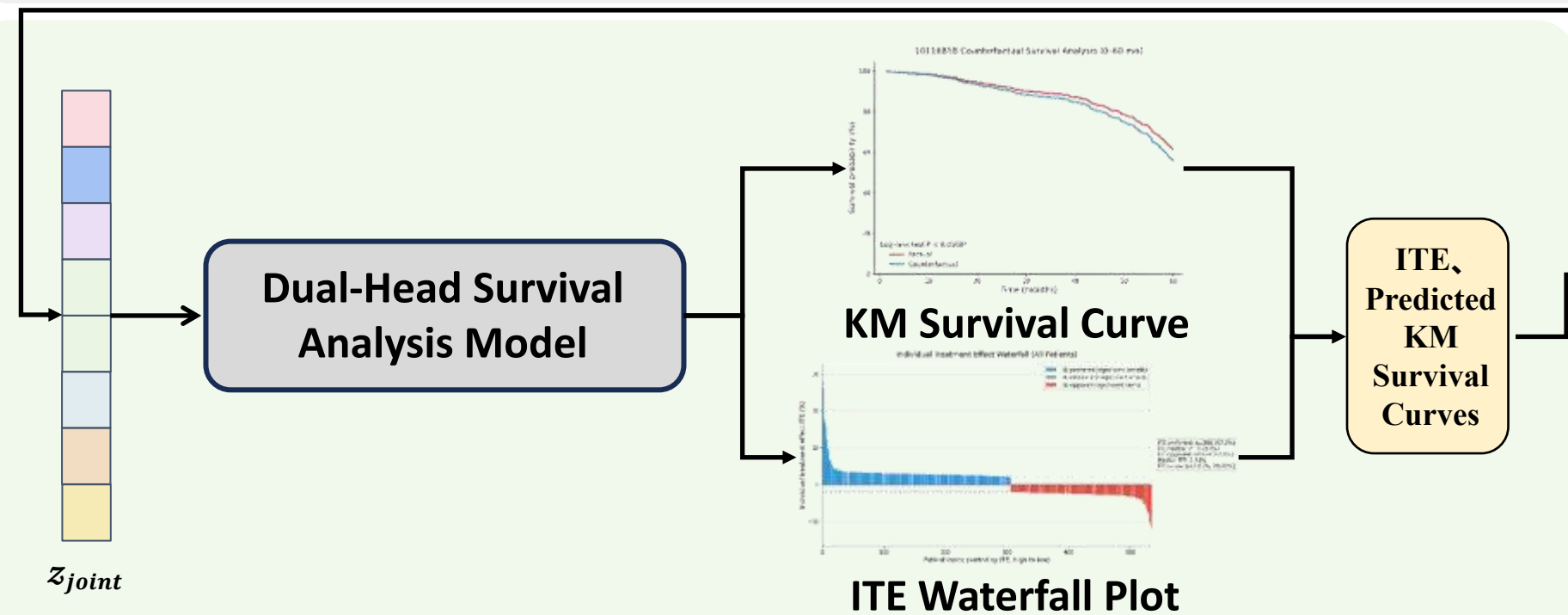
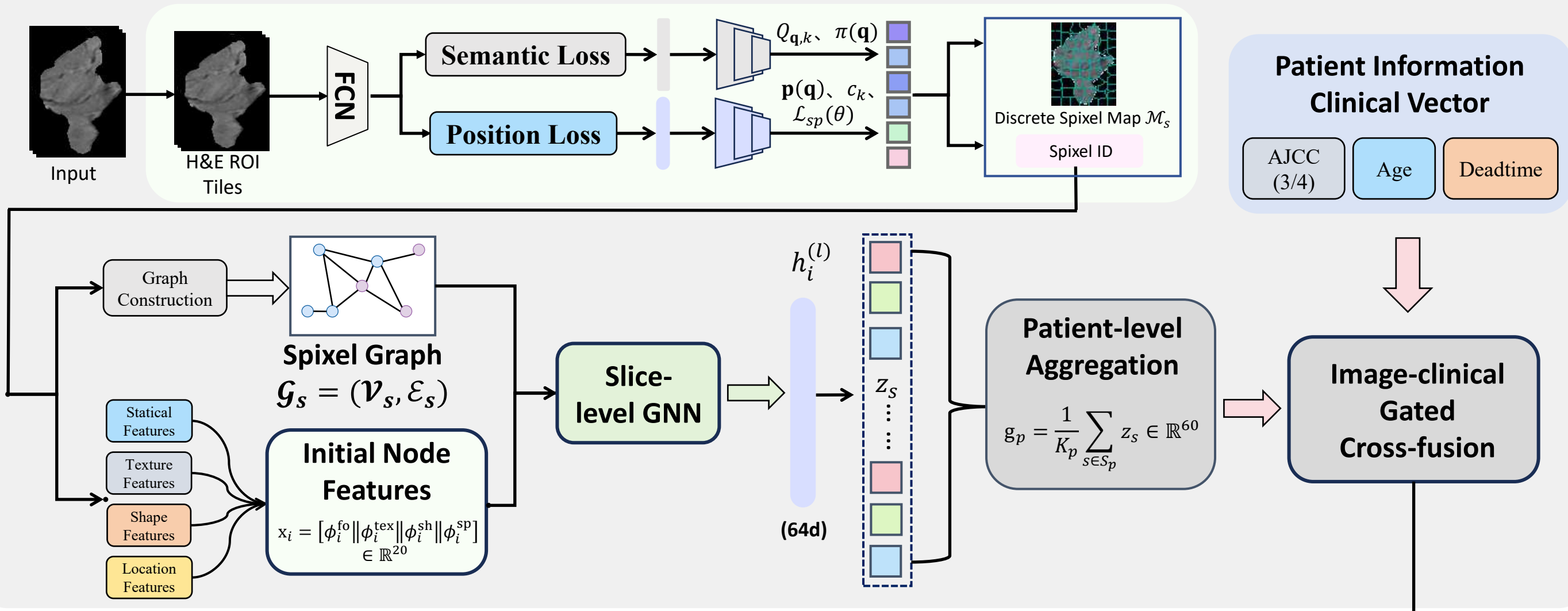
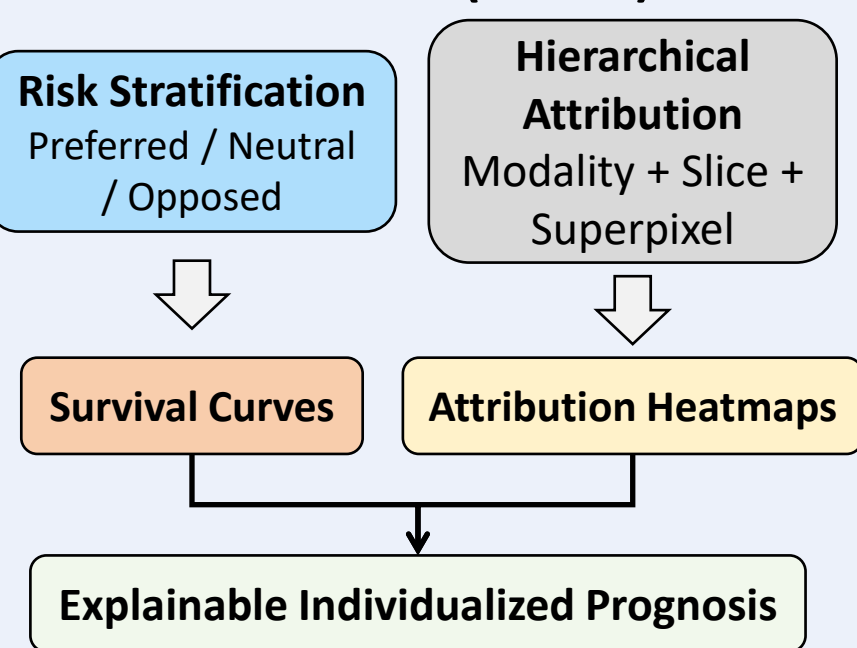


# SP-HAA: Superpixel-to-Patient Hierarchical Embedding and Aggregation



## Risk Stratification and Region-level Attribution (RS-RA)



## DH-CaS: Dual-Head Causal Survival Modeling with Counterfactual Reasoning